

# Learn Path

A full-stack web app designed, built, and deployed across three AI tools in one session

|                                                                   |                                                                             |                                                                        |                                                                       |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>BUILD SESSION</b><br><br><b>12 turns</b><br><br>Claude.ai chat | <b>DEPLOY GUIDE</b><br><br><b>Gemini AI</b><br><br>Step-by-step walkthrough | <b>DEPLOY EXEC</b><br><br><b>Claude Code</b><br><br>Terminal CLI agent | <b>LIVE APP</b><br><br><b>Shipped</b><br><br>learn-path-3d391.web.app |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|

## Part 1 — Build · Claude.ai

|                   |                                                                                                                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Turn 1-2</b>   | <b>Scoped + secured</b><br>Opened with product concept and full tech stack. Before any code, Claude flagged a live API key pasted in chat and proposed the correct architecture: key in a Cloud Function secret, never in the browser.             |
| <b>Turn 3</b>     | <b>Working prototype</b><br>Full React app with glassmorphic UI, AI roadmap generation, and day-checkbox tracker — rendered live and clickable in chat.                                                                                            |
| <b>Turns 4-8</b>  | <b>Five design iterations</b><br>Supplied a UI reference image → rebuilt to glassmorphism + gradient mesh. Iterated: dashboard, project detail, daily log, learn notes with hashtags (#Do/#Don't/#Insight/custom), mobile layout, contrast, fonts. |
| <b>Turns 9-10</b> | <b>Codebase + handoff package</b><br>Generated the full 20-file Firebase project. Then packaged everything into a single markdown document — a prompt-ready bundle handed directly to a second AI for deployment.                                  |

## Part 2 — Deployment guide · Gemini AI

The project package built in Part 1 was handed to Gemini AI, which produced a complete step-by-step deployment plan. Key steps Gemini laid out:

|                                                                                                                                                        |                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Firebase Console</b><br><br>Create project → enable Firestore (asia-southeast1) → enable Anonymous Auth → register Web app and copy firebaseConfig. | <b>Install &amp; link</b><br><br>npm install (root + functions/) → firebase login → firebase use --add to link local code to the Firebase project. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                       |                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environment config</b> <p>Copy .env.example → .env, fill in VITE_FB_ variables from firebaseConfig. Gemini explicitly flagged: do NOT put the Gemini key here.</p> | <b>Secure the key</b> <p>firebase functions:secrets:set GEMINI_KEY — stores it encrypted in Google Cloud Secret Manager, readable only by the backend function.</p>                                   |
| <b>Build &amp; deploy</b> <p>npm run build → firebase deploy. Deploys Firestore rules, Cloud Function (generateRoadmap), and Hosting in a single command.</p>         | <b>Troubleshooting</b> <p>Provided three specific error fixes: Firestore permission denied → check Anonymous Auth; roadmap fails → firebase functions:log; env mismatch → verify VITE_FB_ values.</p> |

Part 3 — Deploy execution · Claude Code CLI

With Gemini's guide as context, Claude Code was run inside the project folder. It diagnosed and resolved every blocker autonomously:

|             |                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diagnose    | Explored project files, detected firebase.json, confirmed Firebase CLI (v15.18) installed.                                                                           |
| Account fix | Detected wrong Google account (nttrong204@gmail.com). Instructed firebase login:add, then switched to thetrongnguyen204@gmail.com and re-linked the correct project. |
| Secret      | Ran firebase functions:secrets:set GEMINI_KEY — confirmed: "Created a new secret version projects/289347860731/secrets/GEMINI_KEY/versions/1".                       |
| Build       | npm run build → Vite compiled React app to dist/ successfully.                                                                                                       |
| Deploy      | firebase deploy → Hosting, Cloud Function, Firestore rules all deployed. App live at https://learn-path-3d391.web.app                                                |
| ■ LIVE      | https://learn-path-3d391.web.app                                                                                                                                     |

What this demonstrates

**Multi-AI orchestration**  
 Used three AI tools in sequence — each for what it does best: Claude.ai for product design, Gemini for structured planning, Claude Code for autonomous execution.

**Security instinct**  
 API key was flagged and secured at two points — once during build (architecture change), once during deploy (Cloud Function secret).

**Specific direction**  
 Every prompt had a concrete brief: font names, reference images, component behaviour. Output quality matched input quality.

**End-to-end ownership**  
 Concept → prototype → codebase → GitHub → live URL. Nothing handed off to a developer. Shipped independently using AI tooling.

